Yaroslav Alekseev

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http://aleksyar.github.io

Education

Technion – Israel Institute of Technology — Haifa, Israel
***Ph.D.** in Computer Science* — 2022 - present
- Advisor: Prof. Yuval Filmus
- Research area: Computational Complexity

Saint-Petersburg State University — St. Petersburg, Russia
M.Sc. in Mathematics — 2020 - 2022
- Advisor: Prof. Edward A. Hirsch
- Research area: Propositional Proof Complexity

Saint-Petersburg State University — St. Petersburg, Russia
B.Sc. in Mathematics — 2016 - 2022
- GPA: 4.96/5.00

Internships

Yandex | St. Petersburg, Russia — 2019.07 - 2019.11
- Data analyst

Industry projects

HuaweiSAT | St. Petersburg, Russia — 2021.11 - 2022.06
- SAT Solver Project: Developing a solver specifically optimized for circuit satisfiability (Circuit-SAT)

Personal projects

Resolution in Lean 4
Formalization of the Resolution proof system

- Formalized the Resolution proof system in Lean 4 and proved several classical results about it.

Awards and Fellowships

- **Best Student Paper award**, Computational Complexity Conference 2021 — 2021
- **Lady Davis Fellowship** — 2022.10-2023.10
- **Faculty Excellence Scholarship** — 2024.10-2025.10
- **7th place at Russian National Math Olympiad** — 2016

Skills

- **C++**
- **Python**
- **Lean 4**
- **SQL**

Languages

- **English**
- **Russian**

PUBLICATIONS

1. **Yaroslav Alekseev**, Dima Grigoriev, Edward A. Hirsch, and Iddo Tzameret. Semi-algebraic proofs, IPS lower bounds and the τ-conjecture: Can a natural number be negative?
   In ACM Symposium on Theory of Computing, 2020 (**STOC 2020**)
2. **Yaroslav Alekseev.** A Lower Bound for Polynomial Calculus with Extension Rule.
   In Computational Complexity Conference, 2021 (**CCC 2021**)
3. **Yaroslav Alekseev**, Edward A. Hirsch. The Power of the Binary Value Principle.
   In International Conference on Algorithms and Complexity, 2023 (**CIAC 2023**)
4. **Yaroslav Alekseev**, Alexander Smal, Yuval Filmus. Lifting Dichotomies.
   In Computational Complexity Conference, 2024 (**CCC 2024**)
5. **Yaroslav Alekseev**, Dima Grigoriev, Edward Hirsch. Tropical proof systems.
   In International Symposium on Theoretical Aspects of Computer Science (**STACS-2025**)
6. **Yaroslav Alekseev**, Dmitry Itsykson. Lifting to regular resolution over parities via games.
   In ACM Symposium on Theory of Computing, 2025 (**STOC 2025**)
7. **Yaroslav Alekseev**, Mika Göös, Ziyi Guan, Gilbert Maystre, Artur Riazanov, Dmitry Sokolov, Weiqiang Yuan. Generalised Linial-Nisan Conjecture is False for DNFs.
   In Computational Complexity Conference, 2025 (**CCC 2025**)
8. **Yaroslav Alekseev**, Yuval Filmus, Ian Mertz, Alexander Smal and Antoine Vinciguerra. Catalytic Computing and Register Programs Beyond Log-Depth.
   In Mathematical Foundations of Computer Science, 2025 (**MFCS 2025**)
9. **Yaroslav Alekseev**, Yuval Filmus. Approximate polymorphisms of predicates.
   Preprint available on ECCC.
10. Aryan Agarwala, **Yaroslav Alekseev**, Antoine Vinciguerra. Linear Matroid Intersection is in Catalytic Logspace.
    In 17th Innovations in Theoretical Computer Science, 2026 (**ITCS 2026**)
11. **Yaroslav Alekseev**, Nikita Gaevoy. Intersection Theorems: A Potential Approach to Proof Complexity Lower Bounds.
    In 17th Innovations in Theoretical Computer Science, 2026 (**ITCS 2026**)
12. **Yaroslav Alekseev**, Mika Göös, Konstantin Myasnikov, Artur Riazanov, Dmitry Sokolov. Sampling Permutations with Cell Probes is Hard. *In ACM Symposium on Theory of Computing, 2026* (**STOC 2026**)
13. **Yaroslav Alekseev**, Nikita Gaevoy. New Polynomial-depth Res(+) Lower Bounds.